

Temporal Anti-Aliasing for Single-Photon Reconstruction

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Single-photon avalanche diode (SPAD) cameras capture high-speed bursts of binary frames, where each pixel records only whether a photon arrived during a given exposure. A single frame carries almost no usable signal; a clean image only emerges from aligning and merging an entire burst (quanta burst photography). This project investigates single-photon image reconstruction under motion, specifically whether history-rejection heuristics from real-time temporal anti-aliasing (TAA), namely neighborhood clamping and motion-adaptive blending, improve classical align-and-merge reconstruction, particularly under fast motion and disocclusion. We propose to evaluate on the Single Photon Challenge benchmark, co-organized by Prof. Gupta’s lab, comparing a TAA-augmented pipeline against naive-averaging and hierarchical block-alignment baselines.

1. Introduction

Single-photon avalanche diode (SPAD) cameras capture high-speed bursts of binary frames, where each pixel records only whether a photon arrived during a given exposure. A single frame carries almost no usable signal; a clean image only emerges from aligning and merging an entire burst, a process known as quanta burst photography. We address **single-photon image reconstruction under motion**: specifically, whether history-rejection heuristics from real-time temporal anti-aliasing (TAA), namely neighborhood clamping and motion-adaptive blending, improve classical align-and-merge reconstruction, especially under fast motion and disocclusion.

This problem matters for two reasons: SPAD arrays are an emerging sensor technology, with megapixel-scale arrays now available, making motion-robust reconstruction increasingly practical; and the standard SPAD pipeline (estimate motion, align, merge) is structurally identical to TAA in real-time rendering, a connection we have not found explored elsewhere.

The project builds directly on Prof. Gupta’s own quanta burst photography research and the Single Photon Challenge benchmark his lab co-organized, giving the team a public dataset with ground truth rather than requiring custom data collection.

2. Related Work

Quanta burst photography (Gupta et al.) establishes the align-then-accumulate pipeline for SPAD bursts that serves as our baseline.

Temporal anti-aliasing. A survey of TAA techniques (Yang et al., NVIDIA, 2020) catalogs the history-rejection methods we adapt to this domain: neighborhood clamping and motion-adaptive blend weights.

Single Photon Challenge. This benchmark provides the dataset and existing leaderboard submissions used for comparison in this work.

State of the art. The current state of the art is the quanta burst photography pipeline: align each binary frame to a reference (block matching or optical flow), then merge by averaging. This works well under static or slow motion, but once alignment becomes unreliable, under fast motion, disocclusion, or low light, misaligned pixels are still blended in at full weight; the pipeline has no mechanism to detect and down-weight an untrustworthy alignment, the failure mode we target.

3. Approach

We first re-implement the existing quanta burst photography baseline as a validated comparison point, then propose a new approach not previously explored to our knowledge: adapting per-pixel history-rejection heuristics from TAA to explicitly reject unreliable alignments rather than blending them in regardless. Concretely:

1. Implement a *baseline*: naive frame averaging, followed by hierarchical block alignment and merging (a simplified quanta burst photography pipeline).
2. Implement a *TAA-augmented pipeline* using the same alignment and merge backbone, adding neighborhood-clamping history rejection, motion-adaptive blend weights, and a per-pixel confidence map.
3. Evaluate both pipelines on held-out Single Photon Challenge training scenes and on synthetic motion sweeps generated with the visionsim simulator.
4. Analyze where and why each method breaks down under fast motion and disocclusion.

Figure 1 illustrates the per-frame mechanism of the TAA-augmented pipeline and the point at which the naive-averaging baseline diverges from it. This approach draws on course material including image alignment and transformations, RANSAC, optical flow, tracking, and the quanta computational imaging lecture.

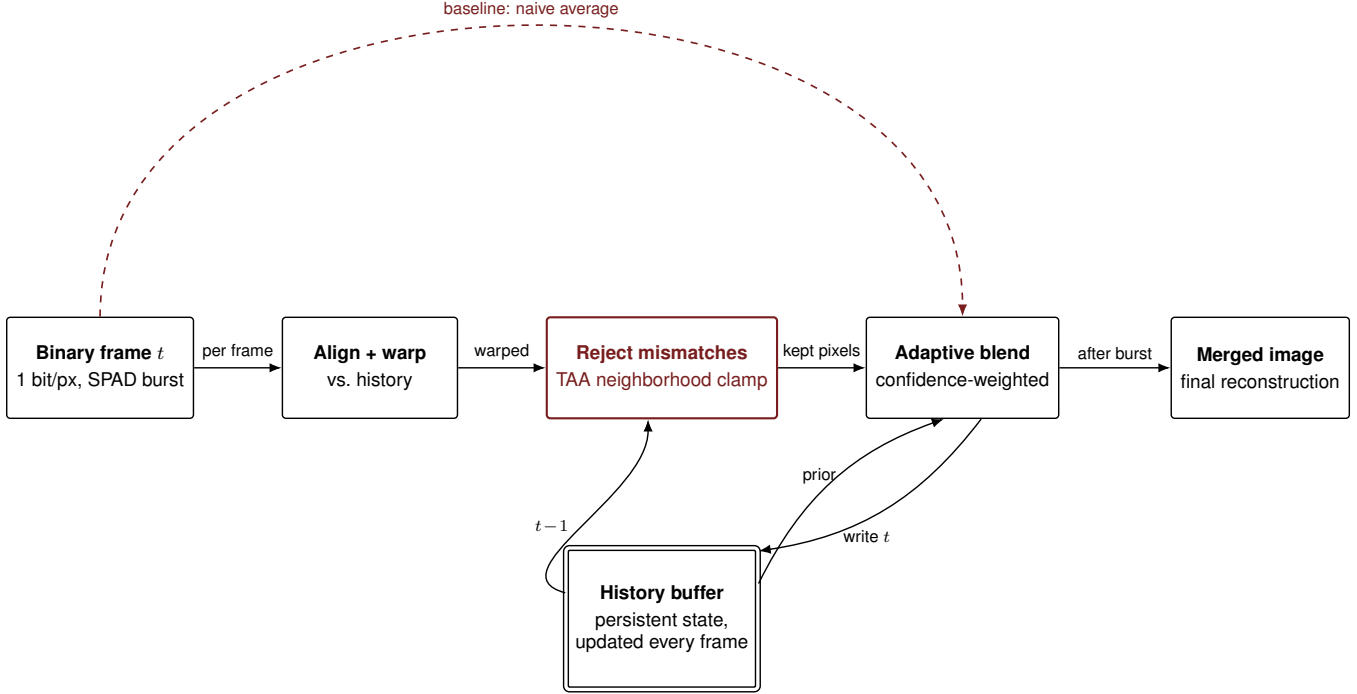


Figure 1: Per frame, the TAA-augmented pipeline (solid) warps the incoming binary frame to align with the running history buffer, rejects pixels where that alignment is unreliable, and adaptively blends the rest back into the history, which feeds the next frame and, after the full burst, the final reconstruction. The naive-averaging baseline (dashed) instead accumulates every frame directly, with no alignment or rejection.

We expect the TAA-augmented pipeline to outperform the baseline specifically in the fast-motion and disocclusion regime, where naive blending is most error-prone. The neighborhood clamp checks whether the aligned history agrees with the incoming frame’s local neighborhood and rejects the pixel when it does not, and the confidence-weighted blend further downweights marginal agreement; the baseline has no analogous mechanism and blends every aligned pixel at a fixed or naively motion-scaled weight regardless of trustworthiness. Since both pipelines share the same alignment and merge backbone, any difference in reconstruction quality can be attributed to the history-rejection step itself.

4. Data

The Single Photon Challenge dataset provides 50 simulated training scenes and 5 test scenes (ground truth withheld), each burst (a “photon cube”) consisting of 1,024 binary frames. The full set is approximately 425 GB; we will work with a 5–10 chunk subset (roughly 8.5 GB per chunk), supplemented with our own synthetic motion sweeps generated via the MIT-licensed visionsim simulator.

5. Evaluation Plan

We will report PSNR and SSIM on held-out training scenes, and conduct a failure analysis against motion speed using visionsim sweeps to characterize when temporal reconstruction breaks down. The contribution is framed as an interpretable

classical method with a rigorous failure analysis, rather than a state-of-the-art claim against neural approaches.

6. Timeline

Wk	Dates	Milestone
1–2	Sep 29–Oct 12	Data pipeline and baseline working
3–4	Oct 13–Oct 26	TAA-augmented pipeline implemented
5	Oct 27–Oct 29	Mid-term report: baseline vs. TAA
6–8	Oct 30–Nov 19	Motion sweeps, failure analysis
9	Nov 20–Nov 30	Final evaluation, ablations
10	Dec 1–Dec 10	Presentation and final webpage

Table 1: Project timeline by course week.

7. Scope and Feasibility

This project topic was discussed with Prof. Gupta, who confirmed it is a reasonable and worthwhile direction. As expected at this stage, scope may be refined as implementation progresses. Prof. Gupta also offered to connect the team with his lab for data access or implementation guidance if needed.